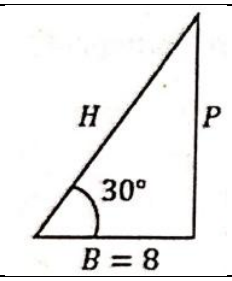
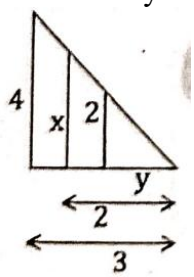
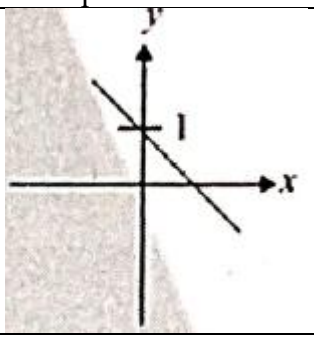
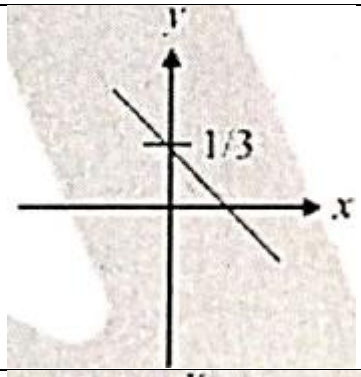
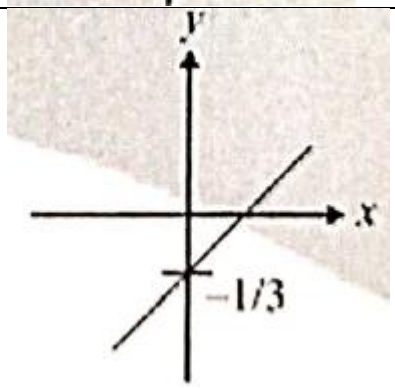
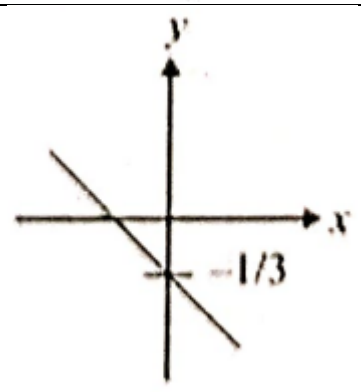
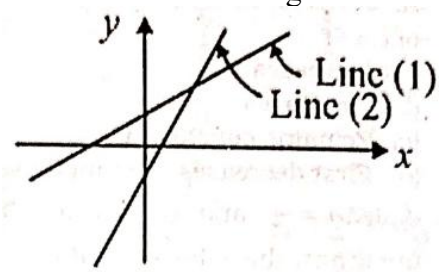
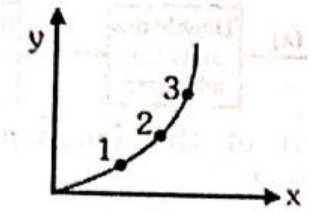
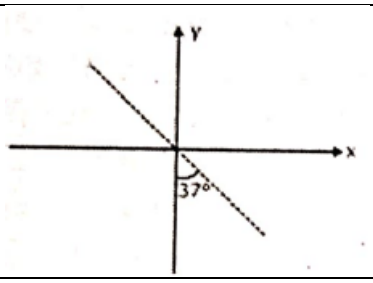
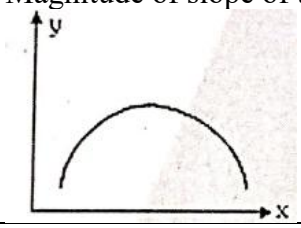
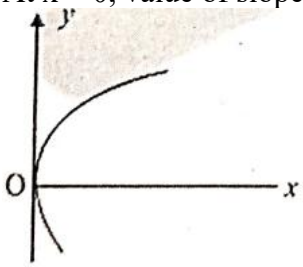
Question	As θ increases from 0° to 90°, the value of $\cos\theta$:-
Type	multiple choice
Option	Increases
Option	Decreases
Option	Remains constant
Option	First decreases then increases.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If $\tan\theta = \frac{1}{\sqrt{5}}$ and θ lies in the first quadrant, the value of $\cos\theta$ is:
Type	multiple choice
Option	$\sqrt{\frac{5}{6}}$
Option	$-\sqrt{\frac{5}{6}}$
Option	$\frac{1}{\sqrt{6}}$
Option	$-\frac{1}{\sqrt{6}}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following option is correct for the value of sin θ.
Type	multiple choice
Option	2
Option	$1/\sqrt{2}$
Option	$\sqrt{2}$
Option	$\frac{\sqrt{5}}{2}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Convert angle from radian to degree, $\frac{5\pi}{6}$ rad:
Type	multiple choice
Option	60^0
Option	30^0
Option	90^0
Option	150^0
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Find the value of P

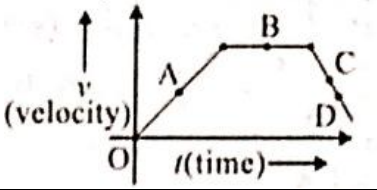
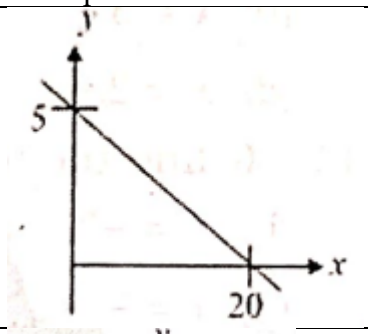
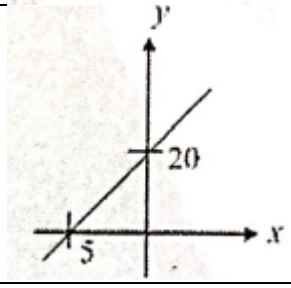
	
Type	multiple choice
Option	$\frac{\sqrt{3}}{8}$
Option	8
Option	$\frac{8}{\sqrt{3}}$
Option	0
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Find x and y: 
Type	multiple choice
Option	$x=\frac{4}{8}, y=\frac{3}{2}$
Option	$x=\frac{8}{3}, y=\frac{6}{2}$
Option	$x=\frac{8}{3}, y=\frac{3}{2}$
Option	$x=\frac{3}{2}, y=\frac{8}{3}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Find the value of sin(15°)
Type	multiple choice
Option	$\frac{\sqrt{3}+1}{2\sqrt{2}}$
Option	$\frac{1-\sqrt{3}}{2\sqrt{2}}$
Option	$\frac{\sqrt{3}-1}{2\sqrt{2}}$

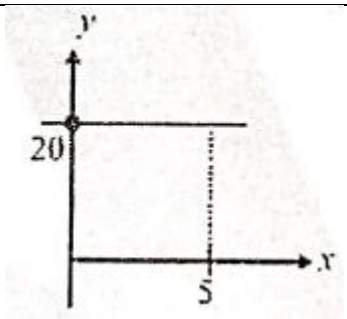
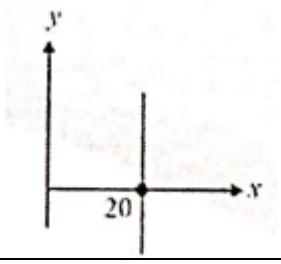
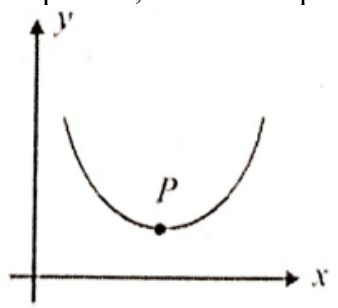
Option	$\frac{\sqrt{3}-1}{\sqrt{2}}$
Answer	3
Solution	4
Positive Marks	4
Negative Marks	1
Question	If $a=\frac{1}{2}$, then,
Type	multiple choice
Option	$a=\sin 30^0$
Option	$a=\cos 30^0$
Option	$a=\tan 30^0$
Option	$a=\operatorname{cosec} 30^0$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle moves along the straight line $y=3x+5$. Which coordinate changes at a faster rate?
Type	multiple choice
Option	x-coordinate
Option	y-coordinate
Option	Both x and y coordinates
Option	Data insufficient.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The equation of line is $2y=3x-6$ its x and y intercepts are
Type	multiple choice
Option	$-3,2$
Option	$3,-2$
Option	$2,-3$
Option	$-2,3$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Correct graph of $3x+3y+1=0$ is
Type	multiple choice
Option	

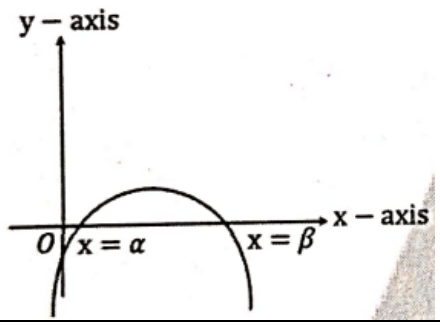
Option	
Option	
Option	
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The equation of line making an angle 135° with the positive x-axis and passing through a point (2,3) will be:
Type	multiple choice
Option	$y-x=5$
Option	$y=x-5$
Option	$y-x=1$
Option	$y=-x+5$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The equation of the line is $3y=-2x+6$ its x and y intercepts are
Type	multiple choice
Option	$-3,2$
Option	$3,2$
Option	$2,-3$
Option	$-2,3$
Answer	2

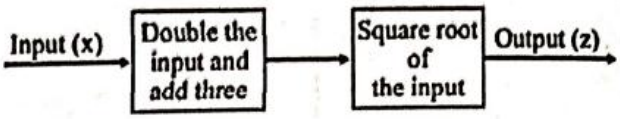
Solution	
Positive Marks	4
Negative Marks	1
Question	Distance between points $(-2,0,-5)$ and $(2,-3,-5)$ is
Type	multiple choice
Option	10
Option	5
Option	12
Option	14
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is not correct for following straight line graph: 
Type	multiple choice
Option	Line (2) has negative y intercept
Option	Line (1) has positive y intercept
Option	Line (2) has positive slope
Option	Line (1) has negative slope
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The slope of graph as shown in figure at points 1,2 and 3 is m_1, m_2 and m_3 respectively then 
Type	multiple choice
Option	$m_1 > m_2 > m_3$
Option	$m_1 < m_2 < m_3$
Option	$m_1 = m_2 = m_3$
Option	$m_1 = m_3 > m_2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Find out the slope of the straight line as shown:-

	
Type	multiple choice
Option	$-\frac{4}{5}$
Option	$-\frac{4}{3}$
Option	$-\frac{3}{4}$
Option	$\frac{3}{4}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Magnitude of slope of the shown graph. 
Type	multiple choice
Option	First increases then decreases
Option	First decreases then increases
Option	Increases
Option	Decreases
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	At x = 0, value of slope is: 
Type	multiple choice
Option	0
Option	1
Option	- 1
Option	Infinity
Answer	4
Solution	
Positive Marks	4

Negative Marks	1
Question	The slope of straight line $\sqrt{3}y = 3x + 4$ is
Type	multiple choice
Option	3
Option	$\sqrt{3}$
Option	$\frac{1}{\sqrt{3}}$
Option	$\frac{1}{3}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The slope of v - t is zero at point 
Type	multiple choice
Option	A
Option	B
Option	C
Option	D
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In which of the following graph slope is'+ 4'.
Type	multiple choice
Option	
Option	

Option	
Option	
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	At point P, value of slope is: 
Type	multiple choice
Option	Zero
Option	Positive
Option	Negative
Option	Infinite
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Given $2x^2+5x-12=0$, find the root of x
Type	multiple choice
Option	$x=\frac{3}{2},-4$
Option	$x=-\frac{3}{2},-4$
Option	$x=\frac{3}{2},-2$
Option	$x=-\frac{3}{2},4$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

Question	The equation $x^2+8x+12=0$ has
Type	multiple choice
Option	No root
Option	One root
Option	Two roots
Option	Four roots
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Find the solutions of given equation $2x^2+3x-2=0$:
Type	multiple choice
Option	$x=-3,\frac{1}{2}$
Option	$x=3,\frac{1}{2}$
Option	$x=-2,\frac{1}{2}$
Option	$x=2,\frac{1}{2}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Given $x^2+7x+12=0,$ find the root of x
Type	multiple choice
Option	$x=\frac{3}{2},-4$
Option	$x=-3,-4$
Option	$x=\frac{3}{2},4$
Option	$x=\frac{3}{2},-2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The following figure shows the graph of $y=ax^2+bx+c.$ Then: 
Type	multiple choice
Option	$a<0$
Option	$a>0$

Option	$a=0$
Option	A is an imaginary number
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	In the given figure, each box represents a function machine. A function illustrates what it does with the input.  Which of the following statements is correct?
Type	multiple choice
Option	$z=2x+3$
Option	$z=2(x+3)$
Option	$z=\sqrt{2x+3}$
Option	$z=\sqrt{2(x+3)}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If α,β are roots of the equation $4x^2+3x+7=0,$ then $1/\alpha+1/\beta$ is equal to
Type	multiple choice
Option	$7/3$
Option	$-7/3$
Option	$3/7$
Option	$-3/7$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Find the value of $\log_{10}10^{35}$
Type	multiple choice
Option	28
Option	32
Option	36
Option	35
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	$\log_{10}(xy)=2,$ find the value of xy
Type	multiple choice
Option	500
Option	300
Option	100
Option	400
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	Which of the following is true, if $a^x = b^y$?
Type	multiple choice
Option	$\log a / \log b = x / y$
Option	$\log a / b = x / y$
Option	$\log a / \log b = y / x$
Option	None of the above
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	What is the value of $\log_{10} 80$, when $\log_{10} 2 = 0.3010$?
Type	multiple choice
Option	3.9030
Option	1.9030
Option	1.6020
Option	None of the above
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	$(1+x)^3$ fond the value, if $x << 1$.
Type	multiple choice
Option	$1+x$
Option	$1-3x$
Option	$1+3x$
Option	$1+3x+3x^2+x^3$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Find the value of $\sqrt{0.98}$
Type	multiple choice
Option	0.99
Option	1.02
Option	0.95
Option	0.90
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	$\log_e 15$ is equal to
Type	multiple choice
Option	$\log_e 3 + \log_e 5$
Option	$\log_e 5 - \log_e 3$
Option	$\log_e 10 + \log_e 5$
Option	$\log_e 10 - \log_e 5$
Answer	1
Solution	

Positive Marks	4
Negative Marks	1
Question	Find out the incorrect statement from below
Type	multiple choice
Option	$\log(6)=\log 1+\log 2+\log 3$
Option	$\log(2+3)=\log(2\times 3)$
Option	$\log_{10} 10=1$
Option	$\log_{10} 1=0$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Differentiate: $f(x)=\frac{1}{x}$, with respect to x
Type	multiple choice
Option	In (x)
Option	$-\frac{1}{x^2}$
Option	$\frac{1}{x^2}$
Option	$\text{In}(x)$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Differentiate $f(x)=\sqrt{x}$, with respect to x
Type	multiple choice
Option	$1/\sqrt{x}$
Option	$\frac{1}{2\sqrt{x}}$
Option	$-1/2\sqrt{x}$
Option	$2\sqrt{x}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Differentiate $f(x)=2\cos(x)-6\sec(x)+3$
Type	multiple choice
Option	$-2\sin x-6\sec x\tan x$
Option	$-2\sin x-3\sec x\tan x$
Option	$-2\sin x-6\sec x\cot x$
Option	$2\sin x-4\sec x\tan x$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Differentiate $y=\tan(x)\sec(x)$ w.r.t x.

Type	multiple choice
Option	$y' = \sec^2(x)\tan^2(x) + \sec^3(x)$
Option	$y' = \sec(x)\tan^2(x) + \sec^3(x)$
Option	$y' = \sec(x)\tan^2(x) + \sec^2(x)$
Option	$y' = \sec(x)\tan^2(x) + \operatorname{cosec}^3(x)$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Differentiate $y = \operatorname{cosec} x \cot x$ w.r.t x
Type	multiple choice
Option	$\operatorname{cosec} x\left(\operatorname{cosec}^2 x + \cot^2 x\right)$
Option	$\operatorname{cosec} x$
Option	$-\operatorname{cosec} x\left(\operatorname{cosec}^2 x + \cot^2 x\right)$
Option	$\operatorname{cosec} x\left(\operatorname{cosec}^2 x - \cot^2 x\right)$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Slope of the curve $y = x^3 - e^x + \cos x$ is
Type	multiple choice
Option	$x^2 + e^x - \sin x$
Option	$3x^2 + e^x - \sin x$
Option	$3x^2 - e^x - \sin x$
Option	$x^2 - e^x + \sin x$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The position of the particle is given by $x = \left(3t^2 - e^t\right)$ in m. The velocity (in m/s) of the particle as a function of time *t* is
Type	multiple choice
Option	$2t - e^t$
Option	$3t + e^t$
Option	$6t - e^t$
Option	$6t^2 - e^t$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1